1.1 Topological Preliminaries

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Definition 0.1. A topology $\mathcal{T}$ in a set X is a collection of subsets of X (called open sets) satisfying the following axioms:

[(1)]the union of any number of open sets is open; the intersection of any finite number of open sets is open; $X, \emptyset$ are open.

A topological space is a set X with a topology $\mathcal{T}$ in X.

$\mathcal{T} = \{\emptyset, X\}$ is the trivial topology.

$\mathcal{T} = \mathcal{P}(X)$ is the discrete topology.

Definition 0.2. For $A \subseteq X$, where $\mathcal{T}$ is a topology in X, then $\mathcal{T}_A$ is an induced topology in A.

$$\mathcal{T}_A = \{U \cap A \mid U \in \mathcal{T}\}.$$

Definition 0.3. In a topological space, a closed set is a complement of an open set.

Definition 0.4. For $E \subseteq X$, $\overline{E}$ is said to be the closure of E, where

$$\overline{E} = \bigcap\{V \mid V \text{ is closed and } V \supseteq E\}.$$

Definition 0.5. A neighborhood of $x \in X$ is any open set containing x.

Definition 0.6. A mapping f from X_1 to X_2 ($X_1 \xrightarrow{f} X_2$) is an assignment to each $x \in X_1$ of an element $f(x) \in X_2$ where X_1, X_2 are topological spaces. (Map and mapping are not the same; a mapping is a function between two topological spaces.)

Definition 0.7. The mapping f is continuous if and only if for every open set O_2 in X_2, the set $f^{-1}(O_2)$ is an open set in X_1. (Note f^{-1} need not be a mapping.)

Definition 0.8. For $x \in X_1$, we say that f is continuous at x if for every neighborhood U of $f(x)$, there exists a neighborhood V of x such that $f(V) \subset U$. (f is continuous if and only if it is continuous at every point of X_1.)

We can show the equivalence of both definitions.

Lemma 0.1. The composition of continuous maps is continuous.

3. *Proof.* If $f\colon X \to Y$, $g\colon Y \to Z$ are continuous.

Goal 1: $(g \circ f)^{-1}(V)$ is open in X where V is any open set in Z.

Goal 2:

$$(g \circ f)^{-1}(V) = f^{-1} \circ g^{-1}(V).$$

$\square$

Example 0.1. [(i)]

1. Every mapping is continuous from X to Y if X has the discrete topology. (I showed the converse is NOT true.)
2. If X is a topological space and Y is a set with the trivial topology, then $f\colon X \to Y$ is continuous.

Definition 0.9. The topological product of X_1, X_2 is $X = X_1 \times X_2$ where the topology in X is generated by a basis

$$\mathcal{B} = \{O_1 \times O_2 \mid O_1 \in \tau_1,\ O_2 \in \tau_2\}.$$

(Definition of basis added to notes, not written here.)

Definition 0.10. If

$$p_1 : X_1 \times X_2 \to X_1 \quad \text{and} \quad p_2 : X_1 \times X_2 \to X_2$$

are such that

$$p_1(x_1 \times x_2) = x_1, \qquad p_2(x_1 \times x_2) = x_2,$$

respectively, then p_1, p_2 are projections of $X_1 \times X_2$.

Lemma 0.2. Projections are continuous, open mappings.

Proof. It suffices to show this for p_1; for p_2 the argument is the same.

For the open map property, observe every open set in X can be written as $U \times V$ where U is an open set in X_1, and V is an open set in X_2.

Now,

$$p_1(U \times V) = U.$$

For continuity, observe

$$p_1^{-1}(U) = U \times X_2$$

where $U \subset X_1$ is open. □

Definition 0.11. A family of subsets $\{V_\alpha\}_{\alpha \in I}$ is a <u>covering</u> if

$$\bigcup_{\alpha \in I} V_\alpha = X.$$

If each V_α is an open subset of X, $\{V_\alpha\}_{\alpha \in I}$ is an <u>open covering</u>, and if $|I| < +\infty$ then $\{V_\alpha\}_{\alpha \in I}$ is a finite covering.

Definition 0.12. A topological space X is **compact** if every open covering of X contains a finite covering. A subset A is said to be compact if A is compact in the induced topology.

Remark 0.1.
1. $\mathbb{R}$ is not compact.
2. If $A \subseteq X$, and $|A| < +\infty$, then A is compact.
3. Let $a, b \in \mathbb{R}$, $a \le b$, then $[a, b]$ is compact in $\mathbb{R}$.

Proof. Refer to Theorem 27.1 and Corollary 27.2 of Munkres. □

Theorem 0.1. Let X be a simply ordered set having the least upper bound property. Then every closed interval in X is compact.

(Check out Prahlad sir's notes on General Topology for an alternate path.)

Proof. Given an open covering of $[a, b]$, say $\mathcal{A}$.

STEP 1: For $x \in [a, b]$ there exists $y \in [a, b]$ and $y > x$ such that $[a, y]$ is covered by at most two elements of $\mathcal{A}$. (Do the case where an immediate successor exists and the case where an immediate successor does not exist.)

STEP 2:

$$C = \{y \in [a, b] \mid [a, y] \text{ is covered by finitely many elements of } \mathcal{A}\}.$$

Observe $C \neq \emptyset$ by applying STEP 1 for $x = a$, and apply the least upper bound property where

$$c = \sup C.$$

STEP 3: Show $c \in C$.

STEP 4: $b \in C$. $\square$

Remark 0.2. **Finite intersection property**: In a topological space X, let $\{F_\lambda\}$ be a collection of closed sets. If the intersection of any finite subcollection is non-empty, then

$$\bigcap_\lambda F_\lambda \neq \emptyset.$$

A space has the finite intersection property if and only if the space is compact.

Theorem 0.2. A closed subset of a compact set is compact.

Proof. Let X be compact, and Y be a closed subset of X.

For any open covering of Y, say $\mathcal{A}$, we construct

$$\mathcal{A} \cup \{X \setminus Y\}$$

as an open covering of X, as $X \setminus Y$ is open.

As X is compact, let $\mathcal{A}'$ be a finite subcover of X.

Now, the finite subcover of Y we get from $\mathcal{A}'$ will not contain $X \setminus Y$, hence it is a finite subcover of $\mathcal{A}$. $\square$

Theorem 0.3. A continuous image of a compact set is compact.

Proof. ✓ $\square$

Theorem 0.4. If X is compact, Y is Hausdorff, and $f\colon X \to Y$ is continuous, then $f(X)$ is closed in Y.

Proof. ✓ □

Definition 0.13. A space X is locally compact if for every $x \in X$, there exists a compact neighborhood of x.

Theorem 0.5. Every compact space is locally compact.

Proof. ✓ □

The converse may not be true.

Example 0.2. $\mathbb{R}$ is locally compact but not compact.

Proof. $\mathbb{R}$ is not compact. □

For $x \in \mathbb{R}$, there exists $(x-\varepsilon, x+\varepsilon)$ as a neighborhood of x, and

$$\overline{(x-\varepsilon, x+\varepsilon)} = [x-\varepsilon, x+\varepsilon]$$

is compact.

Definition 0.14. A homeomorphism is a mapping between two topological spaces X, Y, say f, such that f is a bijective map where f, f^{-1} are continuous mappings.

Example 0.3. Add from C. Chandrashekaran.

Note: All topological spaces are T_1, moving forward, unless mentioned otherwise.

Definition 0.15. Two sets M_1 and M_2 are separated by open sets if there exist open sets O_1 and O_2 containing M_1 and M_2, respectively, such that

$$O_1 \cap O_2 = \emptyset.$$

Definition 0.16. Two sets M_1 and M_2 are separated by a real function if there exists a continuous real function

$$f\colon X \to [0,1]$$

such that $f(x) = 0$ for $x \in M_1$ and $f(x) = 1$ for $x \in M_2$.

Definition 0.17.

- T_1: For any disjoint points, there exists an open set containing one and not the other. Equivalently, all singletons are closed.
- T_2 (Hausdorff): Two distinct points can be separated by open sets.
- T_3 (Regular): For a closed set F and $x \notin F$, $\{x\}$ and F can be separated by open sets.
- T_4 (Normal): Disjoint closed sets can be separated by open sets.

Remark 0.3. $T_4 \Rightarrow T_3 \Rightarrow T_2 \Rightarrow T_1$.

Example 0.4. From the book.

Lemma 0.3. Urysohn's lemma. In a normal space, every two disjoint closed sets can be separated by a real function.

Theorem 0.6. Tychonoff's theorem. A product of compact sets is compact.

There are two approaches to proving this theorem:

1. Zorn's lemma.
2. Transfinite induction.